

Review of Matrices

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Definition 1.1 (Eigenvalues and Eigenvectors).

For a square matrix A , if there exists a non-zero vector $\mathbf{v}$ and a scalar λ such that

$$A\mathbf{v} = \lambda\mathbf{v},$$

then λ is called an eigenvalue of A and $\mathbf{v}$ is called an eigenvector corresponding to λ .

Definition 1.2 (Multiplicity of Eigenvalues).

Suppose λ is an eigenvalue of a matrix A and let the characteristic polynomial of A be $p(x) = \det(A - xI)$. Then we have two types of multiplicity for λ :

- **Algebraic multiplicity** of λ : The number of times λ appears as a root of $p(x)$.
- **Geometric multiplicity** of λ : It is equivalent to any of the following (all of which are equal):
 - The number of linearly independent eigenvectors corresponding to λ .
 - The dimension of the corresponding eigenspace: $\dim(\{\mathbf{v} \mid A\mathbf{v} = \lambda\mathbf{v}\})$.
 - The nullity of $A - \lambda I$: $\dim(\text{Null}(A - \lambda I))$.

Theorem 1.1.

For any eigenvalue λ of a square matrix A ,

$$\text{geometric multiplicity of } \lambda \leq \text{algebraic multiplicity of } \lambda.$$

Proof.

Suppose $A \in \mathbb{R}^{n \times n}$ and λ is an eigenvalue of A with geometric multiplicity k . Since $\dim(\text{Null}(A - \lambda I_n)) = k$, we can pick k orthonormal eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ from $\text{Null}(A - \lambda I_n)$, and then extend this set to an orthonormal basis of $\mathbb{R}^n$ by adding $n - k$ more vectors $\{\mathbf{v}_{k+1}, \mathbf{v}_{k+2}, \dots, \mathbf{v}_n\}$. Define the following matrices:

$$V_1 = \begin{bmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_k \\ | & & | \end{bmatrix} \quad V_2 = \begin{bmatrix} | & & | \\ \mathbf{v}_{k+1} & \cdots & \mathbf{v}_n \\ | & & | \end{bmatrix} \quad V = \begin{bmatrix} \cdots V_1 & \cdots V_2 \end{bmatrix} = \begin{bmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_n \\ | & & | \end{bmatrix}$$

where $V_1 \in \mathbb{R}^{n \times k}$ and $V_2 \in \mathbb{R}^{n \times (n-k)}$, and $V \in \mathbb{R}^{n \times n}$ is an orthogonal matrix. Then we have

$$\begin{aligned} AV_1 &= \lambda V_1, & V^\top AV &= \begin{bmatrix} \cdots V_1^\top \\ \cdots V_2^\top \end{bmatrix} A \begin{bmatrix} \cdots V_1 & \cdots V_2 \end{bmatrix} \\ & & &= \begin{bmatrix} \cdots (V_1^\top AV_1) & \cdots (V_1^\top AV_2) \\ \cdots (V_2^\top AV_1) & \cdots (V_2^\top AV_2) \end{bmatrix} \\ & & &= \begin{bmatrix} \cdots (\lambda I_k) & \cdots (V_1^\top AV_2) \\ \cdots \mathbf{O}_{n-k} & \cdots (V_2^\top AV_2) \end{bmatrix} \in \mathbb{R}^{n \times n} \end{aligned}$$

Thus, the characteristic polynomial of A can be computed as

$$\begin{aligned} p(x) &= \det(A - xI_n) = \det(V^\top (A - xI_n) V) \\ &= \det(V^\top AV - xI_n) \end{aligned}$$

$$\begin{aligned}
&= \det \left(\begin{bmatrix} \dots(\lambda \mathbf{I}_k - x \mathbf{I}_k) & \dots(\mathbf{V}_1^\top \mathbf{A} \mathbf{V}_2) \\ \dots \mathbf{O}_{n-k} & \dots(\mathbf{V}_2^\top \mathbf{A} \mathbf{V}_2 - x \mathbf{I}_{n-k}) \end{bmatrix} \right) \\
&= (\lambda - x)^k \cdot \det(\mathbf{V}_2^\top \mathbf{A} \mathbf{V}_2 - x \mathbf{I}_{n-k})
\end{aligned}$$

Therefore, λ is a root of $p(x)$ with multiplicity at least k , which means the algebraic multiplicity of λ is at least k . □